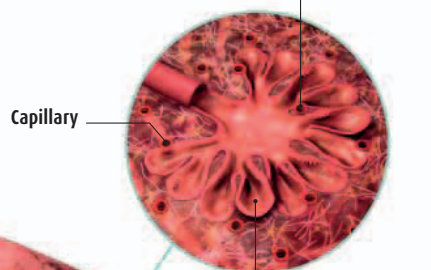
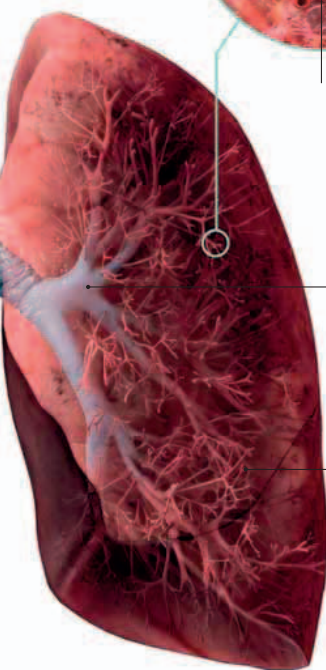


Lungs at 7 weeks The bronchial buds branch out into more buds. Secondary bronchial buds divide to form tertiary buds.

Blood-air barrier The alveoli walls are now so thin that gas can pass through them to the capillaries



Alveoli Terminal sacs are now called alveoli; these continue to form before and after birth



Bronchi These form the main branches of the lungs. Bronchi are joined by fibrous tissue and muscle.

Bronchioles These are made of smooth muscle. They are the terminals of the bronchi.

Lungs at 36 weeks Development of the lungs is usually now complete. The terminal sacs have become thin-walled alveoli that allow gas exchange with the capillaries—a process that transfers oxygen and carbon dioxide between the lungs and bloodstream.

Q At my 20-week ultrasound it looked like my baby was breathing, but how can that be?

Your baby can make breathing-like movements as early as 10 weeks pregnant. These are the result of the involuntary contractions in the muscles that are forming around his lungs. The contractions serve to stimulate more muscle growth and the growth of lung tissue. He begins with short practice bursts that last about 10 seconds, and by week 38 of pregnancy these practice “breaths” are in a regular rhythmic pattern of 40 “breaths” per minute, just as he will breathe after he’s been born. By the time you’re 24 weeks pregnant, your baby will be able to synchronize these breathinglike movements with the rhythmic pumping of his heart, just as he will when he is born.

Q Why doesn’t my baby drown in his amniotic fluid?

The simple answer is that a baby doesn’t drown in amniotic fluid because he doesn’t actually need to breathe when in utero. The important thing for living beings is that we can access oxygen and can remove carbon dioxide from our system, but this gas exchange doesn’t have to occur through breathing—in your baby it happens entirely via the placenta. Your baby’s circulation, which pumps blood around his developing body, actually bypasses his lungs and goes to the placenta for oxygen until the moment of birth. When your baby is exposed to the air after birth, temperature receptors in his skin trigger the absorption of amniotic fluid from the air sacs, the circulation to the placenta “switches off,” and blood starts flowing to his lungs. Then, your baby takes his first breath (which will sound a bit like a gasp), and his lungs will take over the job of breathing. Before the system of bringing oxygen into his body via the lungs is up and running, it isn’t possible for your baby to drown.



Did you know...

Your baby’s lungs will go on developing until he’s about eight years old, forming new bronchial airways and alveoli that will increase his lung capacity, ready for the demands of adolescence and adulthood.

Q How soon is it possible to tell whether my baby is a boy or a girl?

Although your baby’s gender was determined at the moment of conception, it will be at least 14 weeks before there are outward signs. In fact, male and female fetuses look outwardly identical for some weeks (known as the “indifferent” stage) because their genitals are formed from the same structure. This is called the labioscrotal swelling, and it consists of two ridges and a rounded bud. In boys, the ridges connect to form the scrotal sac and the bud elongates to become the penis; in girls, the ridges remain separate and become the labia majora, while the bud shrinks to form the clitoris. By 14 to 17 weeks, the genitals have become obviously differentiated, and have developed enough to be clearly visible.

Internally, the sex organs begin forming at about nine weeks. Depending on whether or not your baby has a Y chromosome (which makes him male), the sex cells create either testes, seminiferous tubules, and vas deferens in your son, or ovaries, uterus, and fallopian tubes in your daughter. The fetal ovaries contain millions of oocytes (immature egg cells)—all the eggs a female will ever have. These could become your future grandchildren. Both boys’ and girls’ sex organs descend from the abdomen to their correct positions after about 25 weeks. The testes have a long journey to make into the scrotal sac, which is why at birth about 1 percent of full-term, and 10 percent of premature male babies have an undescended testis.

At birth, your baby has in the region of 50–70 million alveoli in his lungs, each one supplied with hundreds of tiny blood capillaries ready to exchange oxygen and carbon dioxide.